장기간 무선 모니터링 기반 심근세포 성숙을 위한 전기 자극 최적화

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Optimization of Electrical Stimulation for Cardiomyocyte Maturation Based on Long-Term Wireless Monitoring

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Abstract

The immaturity of in vitro–cultured cardiomyocytes limits their physiological relevance in cardiac modeling and drug evaluation, as indicated by unstable beating rhythms and weak, variable contractile forces. To address this challenge, we developed a multi-channel wireless monitoring platform based on polymer microcantilevers with integrated full-bridge strain sensors for long-term, real-time quantification of cardiomyocyte contraction under programmable electrical stimulation. Continuous monitoring of neonatal rat ventricular myocytes revealed significant temporal fluctuations in spontaneous beating behavior. Controlled electrical stimulation was applied to investigate its role in functional maturation. Long-term force measurements showed that appropriate stimulation enhanced contractile force, improved beating regularity, and reduced beat-to-beat variability compared with non-stimulated controls. Adaptive stimulation strategies further improved contractile stability compared with constant-frequency stimulation. These functional improvements were supported by enhanced sarcomere organization and increased expression of maturation-related markers. This work demonstrates the value of long-term force-based monitoring for optimizing electrical stimulation and promoting cardiomyocyte maturation.

Keywords: ***Cardiomyocyte maturation, Strain sensor, Contractile force, Electrical stimulation***

1. Introduction

Cardiotoxicity remains a major cause of late-stage clinical trial failures and post-market drug withdrawals, emphasizing the need for reliable preclinical screening methods. Cardiomyocyte contractile behavior provides a direct indicator of drug effects and is therefore essential for in-vitro safety assessment. However, commonly used models, including neonatal rat ventricular myocytes and human induced pluripotent stem cell–derived cardiomyocytes, exhibit immature phenotypes characterized by irregular beating, weak contractile force, and underdeveloped sarcomeric organization. This immaturity limits their predictive relevance for adult cardiac responses.

Electrical stimulation has emerged as an effective strategy to promote cardiomyocyte maturation by improving structural organization and functional performance. Yet, most studies rely on fixed-frequency pacing, despite the fact that intrinsic beating behavior and pacing tolerance change dynamically over time. Such mismatches can reduce stimulation efficacy or induce stress.

Furthermore, conventional snapshot measurements fail to capture long-term fluctuations in contractile behavior. Continuous monitoring is therefore essential for rational stimulation design. To address this need, we developed a wireless cantilever-based platform for long-term contractile force measurement, enabling adaptive electrical stimulation optimized for cardiomyocyte maturation[1-3].

2. Experiment and Discussion

To enable rational optimization of electrical stimulation, a wireless multi-channel monitoring system was developed for continuous and noninvasive measurement of cardiomyocyte contractile force. As illustrated in Fig. 1, the platform integrates polymer microcantilevers with on-chip full-bridge strain sensors and a compact wireless acquisition module, allowing long-term, real-time data collection without disturbing the culture environment. This design eliminates the need for wired connections and enables stable recording inside standard incubators over extended periods. Neonatal rat ventricular myocytes were cultured directly on the cantilever surface, and their spontaneous beating behavior was recorded continuously for multiple days to capture the dynamic evolution of cellular function.

Using this system, temporal fluctuations in beating frequency and contractile force were quantitatively characterized (Fig. 2). The data revealed that intrinsic beating dynamics change significantly over culture time, with gradual declines in frequency and variations in contraction amplitude. These observations indicate that conventional fixed-frequency pacing is often mismatched to the evolving functional state of the cells. Based on these insights, an adaptive electrical stimulation protocol was implemented, in which stimulation frequency was automatically adjusted according to the maximum synchronous response of the cardiomyocytes at each stage of culture.

Long-term experiments demonstrated that adaptive stimulation effectively enhanced functional maturation, leading to more regular beating patterns, increased contraction amplitude, and reduced beat-to-beat variability compared with conventional constant-

frequency stimulation (Fig. 3). In addition to functional improvements, structural and molecular analyses confirmed enhanced sarcomere organization and upregulation of maturation-related markers, including increased Connexin 43 expression and elongated sarcomere length. Together, these results validate that continuous force-based monitoring provides a practical and quantitative framework for dynamically optimizing electrical stimulation strategies. The proposed approach offers an effective pathway to improve the physiological relevance of in-vitro cardiomyocyte models for drug screening and cardiac research applications.

Acknowledgments

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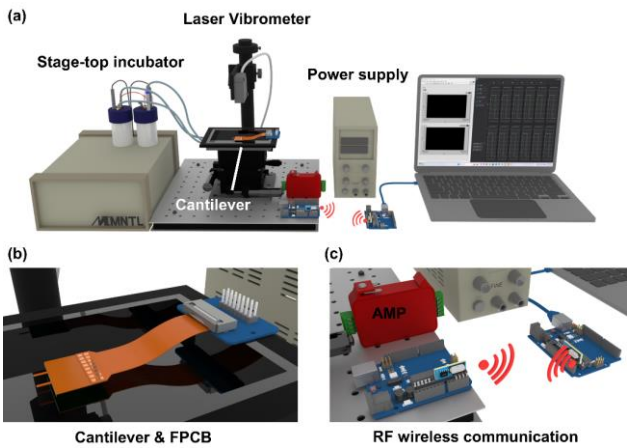

Fig. 1. Schematic of the first-generation wireless monitoring system for cardiomyocyte contraction.

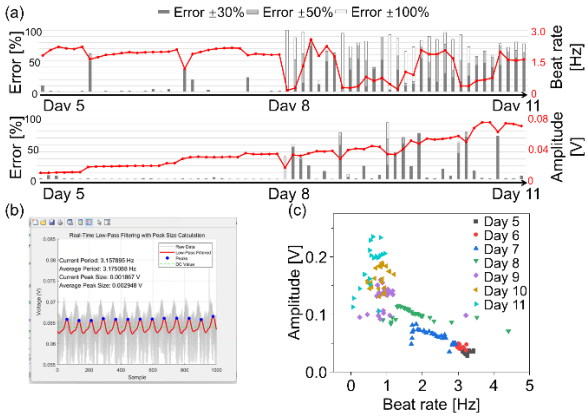

Fig. 2. Long-term analysis of intrinsic cardiomyocyte beating behavior, showing (a) continuous monitoring of spontaneous beating frequency, (b) temporal decline in maximum synchronous response frequency, and (c) variability of beating dynamics over culture time.

Table 1. Electrical stimulation methods.

Day	Without ES	With ES	
	Control (Hz)	Constant(Hz)	Adaptive (Hz)
5	≈ 2.2	1.0	2.5
6	≈ 2.0	1.0	2.3
7	≈ 1.5	1.0	1.8
8	≈ 1.2	1.0	1.5
9	≈ 1.0	1.0	1.3
10	≈ 0.5	1.0	0.8
11	≈ 0.5	1.0	0.8

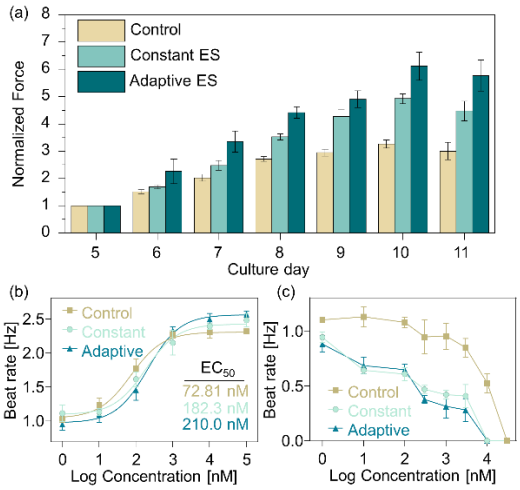

Fig. 3. (a) Evaluation of contraction dynamics under different electrical stimulation (ES). (b) Isoproterenol response. (c) Doxorubicin response.